MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

V SEMESTER BTECH. (E & C) DEGREE END EXAM – Nov/Dec 2022
SUBJECT: MICROWAVE ENGINEERING (ECE - 3154) –SET 1

TIME: 3 HOURS **MAX. MARKS: 50**

Q. No.	Questions	M*	C*	A*	B*
1.	What is mode jumping in Magnetron? Explain various method to eradicate it.	**5**	**2**	**2,3,4,7,8,9,10**	**3**
2.	A reflex klystron is oscillating at 10 GHz when operated in $2\frac{3}{4}$ mode; find the transit time of the electron in repeller space in nano seconds (ns).	**3**	**2**	**2,3,4,7,8,9,10**	**4**
3.	Explain the limitation of conventional tubes at microwave frequencies.	**2**	**2**	**2,3,4,7,8,9,10**	**3**
4.	Draw and explain the Applegate diagram of Reflex klystron	**5**	**2**	**2,3,4,7,8,9,10**	**3**
5.	Two-matched hybrid Tee are connected through two H-plane arms to form a six-port device. If 4 mW of power is fed into port 1 then find the output power in mW in other five ports namely 1', 2, 2', 4, 4'.	**2**	**1**	**1,2,3,4,7,8,9,10**	**4**
6.	Draw current pattern of half wavelength dipole for (i) t=0 (ii)t=T/8,(iii)t=T/4	**3**	**3**	**4,5,7,8,9,10**	**3**
7.	The radiation intensity of an Antenna is $\cos^4\theta$ for the $0 \le \theta \le 90^o$ with $0 \le \Phi \le 360^o$. Find maximum effective aperture using all the formulae for f = 10 GHz.	**5**	**3**	**4,5,7,8,9,10**	**3**
8.	Explain with diagram (a) microstrip, (b) array and (c) Fractal antennas	**3**	**5**	**1,5,6,7,8,9,10**	**3**
9.	What are the challenges for modeling millimetre wave 5G antennas.	**2**	**5**	**1,5,6,7,8,9,10**	**3**
10.	Define the term (a) Polarization Loss Factor (b) Antenna efficiency	**2**	**3**	**4,5,7,8,9,10**	**3**
11.	Starting from fundaments derive the expression for radiation resistance of an infinitesimal dipole	**4**	**4**	**1,5,6,7,8,9,10**	**3**
12.	Derive the value of directivity for a small dipole	**4**	4	**1,5,6,7,8,9,10**	**3**
13.	Starting from fundamentals obtain the value of radiation resistance of a half wave dipole	**4**	4	**1,5,6,7,8,9,10**	**3**
14.	Derive vector wave equation in terms of magnetic vector potential and obtain its solution	**4**	**3**	**4,5,7,8,9,10**	**3**
15.	A resonant half wavelength dipole is made out of copper (σ = 5.7 x 10^7 S/m) wire. Determine the conduction and dielectric (radiation) efficiency of the	**2**	**3**	**4,5,7,8,9,10**	**3**

	dipole antenna at f = 100 MHz if the radius of the wire b is 3 x $10^{-4}\lambda$, and the radiation resistance of the $\lambda/2$ dipole is 73 ohms				

M*--Marks, C*--CLO, A*--AHEP LO, B* Blooms Taxonomy Level